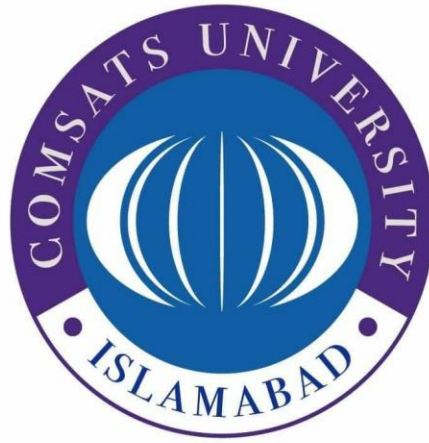




Department of Computer Science,
CUI, Attock Campus.

Program: BSSE

Assignment no:3

Information Security

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Registration #	FA24-BSE-056

Professor: Ms. Ambreen Gull

Session: Fall 2024

Question no1:

Write a Python function to calculate GCD using Euclid's Algorithm.

Input:

```
def gcd(a, b):  
    if b == 0:  
        return a  
    if a == 0:  
        return b  
    return gcd(b, a % b)  
  
# taking input from user  
num1 = int(input("Enter first number: "))  
num2 = int(input("Enter second number: "))  
  
# calling function  
result = gcd(num1, num2)  
  
# printing result  
print("GCD is:", result)
```

Output:

```
PS F:\VS code folder ( IS_ Projects)> & C:\Users\hk\AppData\Local\Programs  
\Python\Python314\python.exe "f:/VS code folder ( IS_ Projects)/2.py"  
Enter first number: 18  
Enter second number: 12  
GCD is: 6  
PS F:\VS code folder ( IS_ Projects)> |
```

Question no2:

GCD with Step Display Modify the GCD function to display each step of Euclid's Algorithm.

Input:

```
def gcd(a, b):
    print("Steps of Euclids Algorithm:")

    while b != 0:
        print("a =", a, ", b =", b, " -> a % b =", a % b)
        a, b = b, a % b

    print("GCD is", a)
    return a

# taking input
num1 = int(input("Enter first number: "))
num2 = int(input("Enter second number: "))

# function call
gcd(num1, num2)
```

Output:

```
PS F:\VS code folder ( IS_ Projects)\3.py
Enter first number: 18
Enter second number: 12
Steps of Euclids Algorithm:
a = 18 , b = 12 -> a % b = 6
a = 12 , b = 6 -> a % b = 0
GCD is 6
PS F:\VS code folder ( IS_ Projects)> |
```
